
AI Agent-Driven Multiscale Ion Bombardment Simulation Framework for Cryogenic Plasma Etching

Gemini 3 flash, Gemini 3 pro, ChatGPT 5.2, GLM-4.7, Claude Opus 4.5, Claude Sonnet 4.5

Abstract

Cryogenic plasma etching relies on complex multiscale phenomena where atomic-level surface interactions dictate macroscopic trench profile evolution. While coupled Molecular Dynamics (MD), Machine Learning (ML), and Monte Carlo (MC) approaches offer a theoretical pathway for predictive modeling, their practical utility is severely limited by fragmented, manual workflows. To address this, we introduce MultiSim, an AI agent-driven framework that autonomously orchestrates the end-to-end multiscale simulation pipeline. The system employs an inspection-guided agent to generate targeted ion-surface interaction data via MD, which are then encoded into a Mixture Density Network (MDN) surrogate and seamlessly integrated into MC trench simulations. Validated on Si, SiO₂, and Si₃N₄ systems, the framework successfully captures material-dependent energy transfer mechanisms, demonstrating the effective, autonomous translation of atomistic physics into device-scale predictions.

1 Introduction

Plasma etching plays a central role in modern semiconductor manufacturing, where nanoscale pattern transfer determines device performance and yield [1]. Understanding and predicting etching behavior requires the integration of physical phenomena occurring across multiple spatial and temporal scales, ranging from atomic-scale plasma-surface interactions to mesoscopic-scale trench profile evolution [2]. In principle, combining simulations at these different scales offers a powerful pathway toward predictive process modeling.

As semiconductor devices scale toward sub-3 nm nodes with increasingly aggressive high-aspect-ratio (HAR) structures, etching processes face fundamental challenges in maintaining profile control and uniformity [3]. Particularly in advanced regimes such as cryogenic etching, ions traveling deep into trenches undergo complex trajectories involving multiple wall collisions, reflection, and implantation events. These ion-surface interactions directly determine local energy deposition patterns, which in turn govern spatially non-uniform etching rates, sidewall damage, and profile distortion [4]. Consequently, a quantitative understanding of how incident ion energy and angle distributions evolve as a function of trench depth and geometry is essential for controlling HAR etching outcomes.

To address these multiscale challenges, a hierarchical simulation strategy is required. At the atomistic scale, molecular dynamics (MD) simulations can resolve the physics of single ion-surface collision events, providing energy-angle distributions (EADs) for reflection, sputtering, and implantation [5]. However, MD simulations are computationally prohibitive for modeling macroscopic trench evolution involving billions of ion impacts. Machine learning (ML) surrogate models, particularly mixture density networks (MDNs), can be trained on MD data to efficiently predict probabilistic collision outcomes [6]. These ML models can then be embedded into Monte Carlo (MC) simulations, which stochastically track ion trajectories and cumulative energy transfer

within realistic trench geometries over experimentally relevant timescales [7]. This three-stage workflow—MD for physics, ML for efficiency, MC for scale—forms the conceptual foundation of multiscale plasma etching simulation [8].

In practice, however, performing these simulations in an integrated and systematic manner remains difficult. Atomistic MD, ML surrogate models, and MC simulations are typically developed and executed as independent components [2]. Researchers must manually coordinate data generation, preprocessing, model training, and cross-scale information transfer. This fragmentation means that constructing and operating multiscale simulation pipelines requires substantial domain expertise and continuous human intervention, creating a bottleneck for research efficiency.

While recent advances in computational modeling have improved the fidelity of individual simulation techniques [2], the overall research workflow remains disjointed. Even with reliable MD models and efficient solvers, there exists no unified mechanism to determine when simulations should be executed, how intermediate results should be utilized, or how modeling decisions should be revised based on outcomes. This lack of orchestration severely limits reproducibility, scalability, and the systematic exploration of complex process conditions.

Recent progress in large language models (LLMs) has introduced the possibility of AI agents capable of reasoning over complex tasks, coordinating tools, and executing multi-step workflows [9]. Rather than operating merely as static code generators, such agents can plan actions, interpret intermediate results, and adapt execution strategies through iterative decision-making [10]. These capabilities suggest a new paradigm for scientific computing, in which simulation workflows themselves can be autonomously managed [11].

We hypothesize that the primary challenge in multiscale plasma etching simulation is not the absence of accurate physical models, but the absence of an autonomous system capable of coordinating heterogeneous simulation modules across scales. Addressing this challenge requires shifting the focus from individual simulation methods to the orchestration mechanism that governs their interaction.

In this work, we propose MultiSim, an AI agent-driven multiscale simulation framework for ion scattering in trench structures, demonstrated specifically within the context of cryogenic plasma etching. At the core of the framework is an LLM Orchestrator that supervises the entire simulation workflow. The agent determines when atomistic MD simulations should be performed, how MD-generated data should be used to train ML surrogate models, and how the trained models should be embedded into MC-based trench evolution simulations. Each component—MD, ML, and MC—is encapsulated as an executable tool, enabling structured coordination through the agent.

By embedding multiscale simulation modules within an agent-based control architecture, the proposed framework enables automated, end-to-end ion scattering simulations. The AI agent manages data flow across scales, executes iterative refinement loops, and reduces the need for manual intervention. This design supports reproducible, extensible, and scalable process modeling, providing a foundation for autonomous multiscale simulation in semiconductor manufacturing.

2 Method

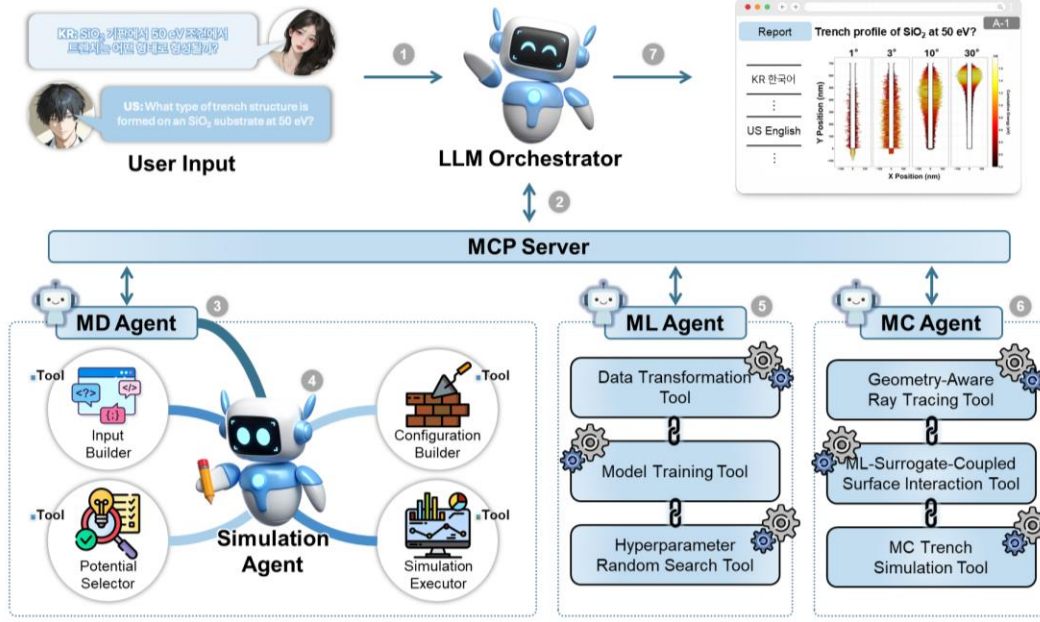


Figure 1. Overview of the LLM-orchestrated AI-agent framework for autonomous multiscale plasma etching simulation.

MD Simulations serve as the foundation for understanding atomic-scale ion-surface interactions. In conventional workflows, MD simulations are performed using packages such as LAMMPS [12] to model single-ion bombardment events on crystalline substrates. Each simulation captures the detailed physics of collision cascades, including ion reflection, implantation, and surface sputtering [5]. By systematically varying incident ion energy (e.g., 0.1-100 eV) and angle (e.g., 0-89°), researchers can generate comprehensive datasets of EADs that characterize surface-response behavior. However, this process requires manual setup of substrate geometries, careful selection of interatomic potentials, parameter specification for each simulation condition, and extensive post-processing to extract statistical distributions from raw trajectory data.

Machine Learning Surrogate Models are then trained to efficiently represent the complex probability distributions extracted from MD data. MDNs have emerged as particularly effective architectures for this purpose, as they can capture the multimodal and stochastic nature of ion-surface interactions [6]. In the conventional approach, researchers manually preprocess MD simulation outputs, extract relevant features (incident energy, angle, and corresponding reflection/implantation outcomes), design network architectures, and train models through iterative hyperparameter tuning. The trained MDN learns to predict probability density functions for reflected ion energy and angle given incident conditions, effectively compressing thousands of MD simulations into a computationally efficient probabilistic model [6].

MC Simulations utilize the trained surrogate models to predict mesoscale trench evolution. In this stage, incident ions are stochastically generated at the trench entrance, and their trajectories are tracked through the three-dimensional geometry [13]. When an ion collides with a trench surface, the MDN model is queried to predict the reflection behavior, and the ion either continues propagating or becomes implanted. Energy deposition at each surface location is accumulated over millions of ion impacts, providing spatially resolved maps of bombardment intensity. This enables prediction of non-uniform etching profiles, sidewall damage distributions, and aspect-ratio-dependent etching phenomena characteristic of HAR structures [4].

While this three-stage workflow has demonstrated scientific validity in prior studies [14], it suffers from the fragmentation and manual coordination challenges outlined in Section 1. To address these limitations, we developed an AI agent-driven multiscale simulation framework. The overall architecture is illustrated in Fig. 1. The framework consists of three specialized agents—(i) an MD simulation agent for atomistic data generation, (ii) an ML surrogate modeling agent for learning surface-interaction behavior, and (iii) an MC trench evolution agent for mesoscale profile prediction—coordinated by a planner LLM through standardized tool interfaces.

2.1 MD simulation agent for inspection-guided atomistic data generation

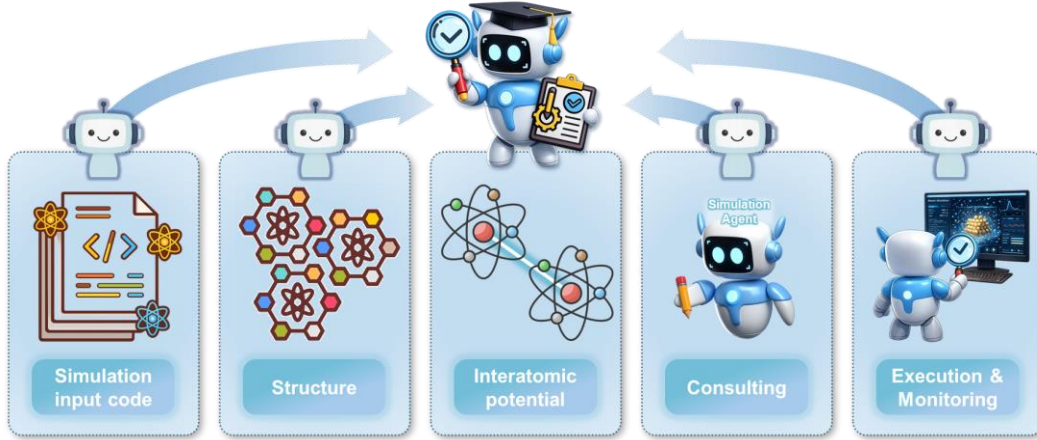


Figure 2. Inspection-guided MD simulation workflow showing the role of the Inspection Agent in validating simulation input code, structure construction, interatomic potential selection, and execution monitoring.

The MD simulation agent generates ion-surface interaction data under the supervision of a dedicated inspection LLM. The agent operates within a multi-LLM architecture in which planning, execution, and verification are explicitly separated [2].

Upon receiving high-level instructions from the Orchestrator, the MD simulation agent begins by retrieving crystallographic information. Given a material identifier (e.g., "Si", "SiO₂"), the agent queries the Materials Project API [15] to obtain crystallographic information files (CIF).

Using the retrieved CIF data, the agent constructs substrate geometries through the Atomic Simulation Environment (ASE) [16]. Unit cells are replicated to achieve sufficient substrate dimensions, and vacuum regions are constructed along the surface-normal direction. Once the substrate is prepared, incident ion configurations are generated based on specified bombardment conditions. The agent constructs molecule files defining projectile atoms with initial positions, velocities corresponding to target kinetic energies, and incident angles sampled according to the desired angular distribution.

With both substrate and projectile configurations established, the inspection agent evaluates the chemical composition and selects appropriate interatomic potentials from available force-field libraries [17]. Physical validity is verified by checking element compatibility, potential parameterization ranges, and suitability for collision dynamics.

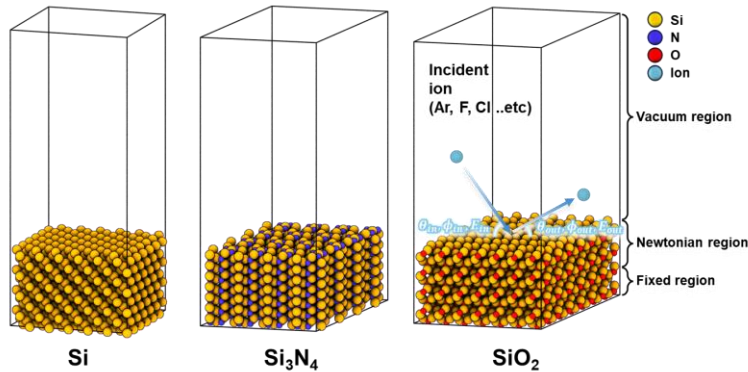


Figure 3. Atomistic substrate structures generated by the MD agent for Si, Si₃N₄, and SiO₂. Incident ions were introduced toward the substrate surface with randomly sampled incidence angles.

Following validation, LAMMPS input scripts are generated using parameterized templates. The

inspection agent performs consistency checks on simulation parameters, including time steps, thermostat settings, and output configurations [18]. Once verifications are complete, MD simulations are executed in local or batch-based computing environments. Figure 3 illustrates examples of constructed simulation systems for different substrate materials (Si, Si₃N₄, SiO₂), showing the substrate structure, vacuum region, and incident ion configuration. Atomistic impact events are recorded as training data for the subsequent machine-learning surrogate model. For each bombardment, incident conditions (energy, angle) and corresponding outcomes (reflection energy and angle, or implantation) are extracted and recorded.

Throughout this workflow, the inspection LLM evaluates physical validity and simulation consistency at each stage, verifying stoichiometry, force-field suitability, geometric parameters, and input-script correctness.

2.2 ML-based surrogate modeling of surface interactions

Surface interaction behavior required for trench-scale simulations is represented using a data-driven surrogate model. Rather than prescribing analytical reflection functions or predefined probabilistic rules, the surrogate model learns statistical surface-response characteristics directly from simulation-generated data.

To capture the intrinsically stochastic nature of plasma-surface interactions, a MDN is employed as the surrogate model. The MDN formulation is particularly suitable for plasma-surface interactions, where multiple surface-response pathways may coexist under identical incident conditions. The MDN learns conditional probability distributions of surface responses given incident particle properties, allowing multimodal and non-deterministic behavior to be expressed within a unified modeling framework.

The surrogate model receives incident particle descriptors, such as energy and incident direction, and predicts the corresponding probability distributions of reflected particle properties. During inference, surface-response events are generated by sampling from the learned distributions, enabling seamless integration with higher-scale simulations.

2.3 MC-based trench evolution driven by ML inference

Trench-scale geometry evolution is simulated using a MC approach [6]. The simulation domain is discretized into surface segments representing trench sidewalls and bottom region.

Incident particles are propagated into the trench geometry, and interaction events are evaluated when particles collide with local surface segments. For each event, surface-response behavior—including reflection and energy deposition—is determined through queries to the ML surrogate model.

Local interaction statistics accumulated on each surface segment are used to estimate etching tendencies, and trench geometry is progressively updated. Through this formulation, the MC simulation incorporates atomistic-level statistical information learned by the surrogate model while operating at mesoscopic length and time scales.

2.4 MCP-based orchestration and planner LLM

All simulation modules are connected through a Model Context Protocol (MCP)-based interface, which provides standardized communication between the planner and executable tools [13]. This interface abstracts internal model implementations and enables the planner to interact with heterogeneous modules in a uniform manner.

Each agent follows workflow execution by selecting appropriate tools, managing intermediate data, and determining when model training, inference, or trench evolution should be performed. By relying on structured tool calls rather than direct code execution, the planner remains decoupled from numerical solvers while retaining full control over workflow logic [11].

This MCP-based orchestration layer enables autonomous execution of MD-ML-MC simulations and supports reproducible and extensible multiscale process modeling.

3 Results and Discussion

3.1 Autonomous MD simulation driven by LLM planning

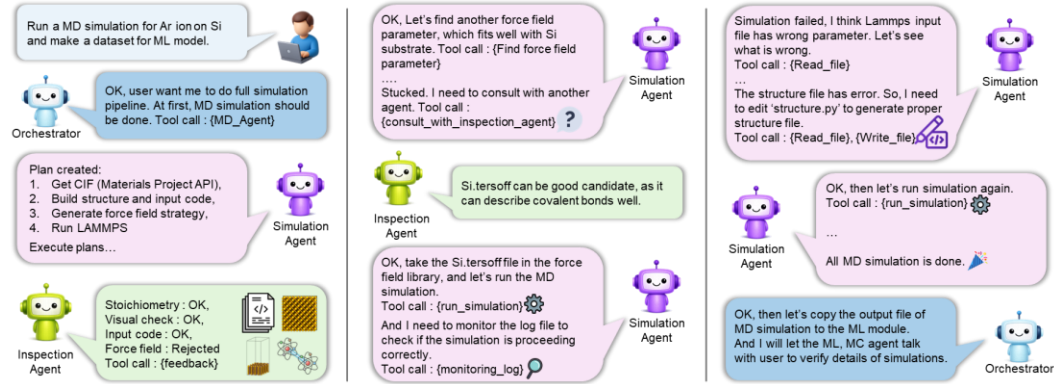


Figure 4. Illustration of an interactive agent-based workflow for autonomous molecular dynamics simulation setup and execution.

First, we evaluate whether the proposed agent-based framework can successfully generate physically consistent atomistic simulation data without human intervention.

Figure 4 illustrates the operational workflow of the MD simulation agent. In this module, given natural language instructions, the Orchestrator autonomously constructed surface-projectile configurations and executed molecular dynamics simulations through the MD simulation agent. The inspection LLM continuously verified structural validity, force-field consistency, and simulation parameter integrity prior to execution [Ref: LLM verification systems].

As a result, stable plasma-surface interaction trajectories were obtained across a wide range of incident energies and angles. The generated trajectories exhibited physically reasonable behaviors, including energy-dependent penetration depth, angular scattering, and surface interaction patterns [15]. These results indicate that the proposed inspection-guided MD agent can reliably produce atomistic training data suitable for downstream surrogate modeling.

3.2 Agent-driven ML-MC simulation workflow

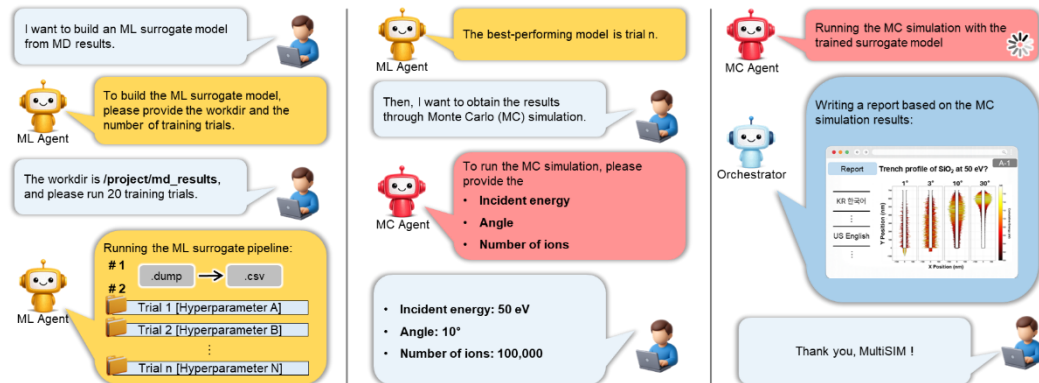


Figure 5: Illustration of the interactive ML-MC simulation workflow enabled by the proposed agent framework.

Surface interaction behavior required for trench-scale simulations is represented using a data-driven surrogate model. Rather than prescribing analytical reflection functions [15] or predefined probabilistic rules [5], the surrogate model learns statistical surface-response characteristics directly from MD-generated data.

Figure 5 illustrates the ML surrogate modeling workflow. Upon receiving a request to build an ML model, the Orchestrator instructs the ML Agent to construct the modeling pipeline. The agent identifies the data source, executes hyperparameter exploration across multiple trials [19], and selects the best-performing model [8]. Once training is complete, the surrogate model is deployed for integration into MC simulations. To capture the intrinsically stochastic nature of plasma-surface interactions, a MDN is employed as the surrogate model. The MDN learns conditional probability distributions of surface responses given incident particle properties, allowing multimodal and non-deterministic behavior to be expressed within a unified modeling framework [20].

The surrogate model receives incident particle descriptors, such as energy and incident direction, and predicts the corresponding probability distributions of reflected particle properties. During inference, surface-response events are generated by sampling from the learned distributions, enabling integration with higher-scale simulations.

3.3 MC trench evolution driven by ML inference

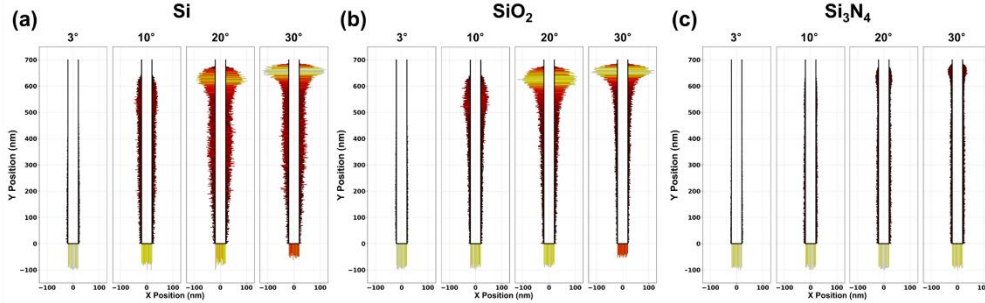


Figure 6. Monte Carlo-based trench energy distribution for (a) Si, (b) SiO₂, and (c) Si₃N₄ substrates at an incident energy of 50 eV and angles of 3°, 10°, 20°, and 30°, and a total of 100,000 ions simulated.

Fig. 6 represents Monte Carlo-based trench energy distributions for Si, SiO₂, and Si₃N₄ substrates obtained using the trained ML surrogate model. All simulations were performed under identical incident ion conditions with an energy of 50 eV, incident angles of 3°, 10°, 20°, and 30°, and a total of 100,000 ions.

Despite identical geometric and injection conditions, distinct material-dependent energy deposition behaviors are observed. For Si (Fig. 6a), pronounced lateral spreading of ion trajectories leads to strong energy accumulation along the trench sidewalls, particularly at higher incident angles [20]. This behavior indicates frequent reflection events and enhanced angular scattering within the trench geometry. For SiO₂ (Fig. 6b), the overall penetration depth is reduced compared to Si [4], and energy deposition becomes increasingly localized toward the upper trench region. While angular scattering remains observable at oblique incidence, sidewall interaction intensity decreases with depth, resulting in attenuated energy transfer to the trench bottom. In contrast, Si₃N₄ (Fig. 6c) exhibits highly directional energy transport with minimal lateral dispersion across all incident angles. Energy deposition is concentrated near the initial impact region, indicating adsorption-dominated surface interactions and limited secondary collision processes.

These results demonstrate that material-specific plasma-surface interaction characteristics learned at the atomistic scale are preserved through ML inference and consistently manifested in mesoscopic trench-scale energy deposition patterns.

3.4 Implications for autonomous multiscale simulation

The results demonstrate several important implications of the proposed agent-driven framework. First, the framework enables effective coupling between atomistic MD simulations and mesoscale MC modeling without requiring manual intervention. The LLM-driven orchestration facilitates seamless data transfer across simulation scales, allowing workflow decisions to be made dynamically rather than relying on rigid, predefined execution scripts.

Second, the observation of physically interpretable trends—such as angle- and energy-dependent etching behavior—indicates that essential physical causality is preserved across scales, despite the use of data-driven surrogate models at intermediate stages.

Finally, the agent-based architecture provides a flexible and extensible foundation for future developments, including adaptive MD sampling strategies, uncertainty-aware surrogate modeling, and closed-loop optimization of plasma processing conditions.

4 Conclusion

In this work, we proposed an AI agent-driven multiscale simulation framework for plasma etching that autonomously orchestrates atomistic MD simulations, machine-learning surrogate modeling, and KMC-based trench evolution.

By introducing an LLM-based planner and inspection-guided execution architecture, the proposed framework enables physically consistent atomistic data generation, data-driven learning of surface-response behavior, and systematic transfer of atomistic interaction statistics to mesoscale geometry evolution. The results demonstrate that angle- and energy-dependent plasma-surface interaction characteristics learned at the atomistic scale can be preserved and reflected in trench-scale etching behavior through the proposed agent-driven pipeline.

Unlike conventional multiscale approaches that rely on manually scripted workflows [5], the present framework emphasizes autonomous decision-making, modular tool coordination, and scalable workflow management. This design significantly reduces human intervention while improving reproducibility and extensibility of multiscale plasma process simulations.

Although several limitations remain—such as dependence on MD data quality and generalization of surrogate models—the proposed framework provides a foundational step toward autonomous, physically grounded multiscale simulation systems. Future extension incorporation adaptive sampling, uncertainty-aware modeling, and closed-loop optimization may further enable intelligent exploration and optimization of plasma etching processes.

5 Limitation of the proposed framework

Despite the advantages of the proposed agent-driven multiscale framework, several limitations remain.

First, the overall performance of the framework is strongly dependent on the quality and representativeness of the atomistic MD simulation data. Since the MD-generated interaction statistics directly serve as training inputs for the ML surrogate model, any bias, insufficient sampling, or modeling error at the MD level can propagate to downstream ML inference and KMC-based trench evolution results. This error propagation across scales remains an inherent challenge in tightly coupled multiscale pipelines.

Second, although the MD simulation agent automates structure construction and input preparation, its ability to handle highly complex plasma-surface phenomena remains limited. Processes involving rare reaction pathways, long-time chemical aging, or strongly coupled electronic effects are difficult to resolve within classical MD frameworks and cannot be fully captured by the current agent configuration.

Third, the ML surrogate model learns statistical surface-response behavior based on the available MD dataset, and its predictive capability may degrade when applied to incident conditions outside the training distribution. While the MDN formulation enables multimodal response modeling, systematic uncertainty quantification and active learning strategies are not yet incorporated.

Fourth, the current framework focuses on autonomous orchestration of simulation workflows rather than real-time decision-making. Iterative MD-ML-KMC refinement cycles introduce computational overhead, which may limit applicability in large-scale process optimization or time-sensitive industrial environments.

Finally, the simulation results generated by the proposed framework have not yet been validated against experimental etching data. While the observed trends—such as energy- and angle-dependent profile evolution—are qualitatively consistent with known plasma-surface interaction physics, quantitative comparison with real trench profiles, etch rates, and surface morphologies obtained from plasma etching experiments remains necessary to establish predictive accuracy. This experimental validation is essential for assessing whether the atomistic-to-mesoscale information transfer captures the essential physics governing actual semiconductor manufacturing processes.

Addressing these limitations will require future integration of adaptive MD sampling strategies,

uncertainty-aware surrogate modeling, and tighter feedback mechanisms between scales. Nevertheless, the present framework provides a foundation step toward autonomous and physically grounded multiscale plasma etching simulation.

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AI Co-Scientist Challenge Korea Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction clearly state the paper’s primary contribution as an AI agent-driven orchestration framework for MD–ML–MC multiscale plasma etching simulations, rather than claiming new physical models. These claims are directly supported by the proposed architecture and experimental demonstrations presented in Sections 1–3 and Figures 1–6.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The paper includes a dedicated limitations section that explicitly discusses constraints related to MD data quality, surrogate model generalization, computational overhead, and lack of experimental validation (Section 5). The scope of applicability and potential failure modes of the framework are clearly acknowledged.

3. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The manuscript does not present formal theoretical theorems or analytical proofs, as its primary contribution lies in system architecture design and empirical multiscale simulation workflow demonstration rather than theoretical analysis.

4. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The paper describes the complete simulation pipeline, including MD data generation procedures, ML surrogate model construction, and MC-based trench simulation settings (Sections 2.1–2.4). Key experimental conditions such as incident energies, angles, materials, and ion counts used in the reported results are explicitly specified (Section 3.3, Figure 6).

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: At the time of submission, the data and code are not yet released in an anonymized public repository. However, the manuscript provides detailed methodological descriptions, and an anonymized release with execution instructions is planned for the final version.

6. Experimental Setting/Details

Question: Does the paper specify all the training and test details (e.g., data splits, hyper-parameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: The experimental workflow, including MD simulation condition ranges, surrogate model type (MDN), training procedure, and MC simulation configuration, is described in Sections 2.1–2.3. The evaluation scenarios and operating conditions are clearly stated in the Results section (Section 3).

7. Experimental Statistical Significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: The current results focus on representative trench energy distributions and do not include error bars or confidence intervals. Repeated-run statistical analysis was omitted due to the high computational cost of large-scale MC simulations, and will be considered in future work.

8. Experimental Compute Resources

Question: For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce the experiments?

Answer: [No]

Justification: While the execution workflow and simulation stages are described, detailed information regarding hardware specifications, runtime, and memory usage is not fully reported in the current manuscript. These details will be added in an appendix in a revised version.

9. Code Of Ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: The study involves no human subjects, personal data, or sensitive information and adheres to responsible machine learning research practices. The manuscript emphasizes transparency, reproducibility considerations, and explicit discussion of limitations (Sections 4 and 5).

10. Broader Impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The proposed framework may positively impact semiconductor manufacturing by improving reproducibility and automation of physics-based process modeling (Sections 1 and 4). Potential risks, such as misuse outside validated simulation regimes or overreliance on surrogate predictions, are acknowledged through explicit scope and limitation discussions (Section 5).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper does not release pretrained generative models or high-risk datasets that would require controlled access or additional safeguards. If future releases occur, usage guidelines and scope restrictions will be provided.

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Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

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Justification: While the manuscript cites external tools and software packages (e.g., LAMMPS, ASE, Materials Project), explicit license information and version numbers are not yet fully listed. These details will be added in an appendix in the revised manuscript.

13. New Assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

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Justification: The paper does not introduce or publicly release new datasets or pretrained models as standalone research assets. The primary contribution is a framework and workflow design.

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15. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

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